

Multi-Task Learning for Uzbek News Text Classification

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Abstract—We explore multi-task learning for Uzbek news text classification in a low-resource setting. We introduce a new dataset of more than sixteen thousand Uzbek news articles collected from Kun.uz, annotated for three supervised tasks: topic classification, author gender identification, and publication year prediction. Using a shared-encoder, multi-head architecture based on a pretrained Uzbek transformer model, we compare single-task baselines with several multi-task learning variants, including layer-wise aggregation and cross-task attention. Experimental results show that transformer-based models outperform classical baselines across all tasks. Multi-task learning produces generally stable performance, with modest gains for topic classification and consistent results for gender and temporal prediction. Our analysis using confusion matrices and shared-representation visualizations shows that, in Uzbek news classification under low-resource conditions, task difficulty is mainly determined by how well the labels align with the underlying semantic representations, rather than by architectural complexity. Our study provides the first systematic evaluation of multi-task learning for Uzbek news classification and contributes an annotated dataset and empirical insights for future low-resource natural language processing research.

Keywords: *Uzbek language, news text classification, multi-task learning, low-resource NLP, pretrained language models.*

I. INTRODUCTION

Online news platforms publish large volumes of articles on a daily basis, covering diverse topics and events. Organizing this content in a structured way is essential for navigation, search, and large-scale analysis. Text classification provides an automatic solution by assigning informative labels to news articles, enabling efficient management and analysis of continuously growing news collections.

Text classification for low-resource languages has received increasing attention as multilingual pretraining and transfer learning continue to advance. Despite this progress, high-quality annotated corpora for Central Asian languages remain limited, including for Uzbek, where only recent efforts have produced sentiment datasets, NER resources, and parallel corpora [10], [12], [17]. This scarcity constrains the development of robust natural language processing (NLP) systems and limits empirical research on multi-attribute text inference or representation learning for the Uzbek language. Online news text, in particular, encodes rich linguistic and socio-cultural information—such as topical structure and temporal context—that remains underexplored in Uzbek NLP.

Pretrained language models learn general linguistic representations from large collections of unlabeled text and have become a foundation of modern natural language processing [16]. Recent developments in such models have enhanced representation learning for low-resource languages, making practical text classification possible even when annotated data are limited.

In this work, we investigate multi-task learning (MTL) as a strategy to improve text classification performance for Uzbek news data. MTL leverages related auxiliary tasks to encourage models to learn more generalizable representations, reducing overfitting and improving robustness in low-resource environments. Prior research shows that shared-encoder MTL architectures can enhance sample efficiency and stabilize optimization, especially when tasks provide complementary linguistic or stylistic signals [4], [6]. Our dataset contains three supervised prediction tasks—topic classification, author gender identification, and publication year estimation—each capturing distinct linguistic dimensions that can act as mutually beneficial inductive biases.

We develop a shared-encoder, multi-head architecture and evaluate its effectiveness relative to single-task baselines. Beyond performance comparisons, we conduct detailed analyses of task interactions and shared representations to better understand multitask behavior. Our experimental results show that jointly learning sociolinguistic attributes improves downstream performance across tasks. To our knowledge, this is the first study to apply multi-task learning to Uzbek news text classification, contributing both a modeling approach and an annotated resource for future research in low-resource Uzbek NLP.

II. RELATED WORK

Low-resource NLP for Central Asian languages has progressed unevenly across Uzbek, Kazakh, Kyrgyz, and Turkmen, with Uzbek and Kazakh benefiting from recent dataset creation and benchmarking while Kyrgyz and Turkmen remain particularly underserved. For Uzbek, curated datasets now exist for sentiment analysis and NER, supporting reproducible baselines and facilitating transfer to modern neural architectures. Kuriyozov et al. introduced annotated sentiment corpora (manual and translated) and showed that CNN and classical

models with fastText can be competitive under data scarcity, establishing early baselines and data creation practices for low-resource languages (e.g., translation-aided labeling) [10]. Mengliev et al. released a manually annotated Uzbek NER corpus with BIOES labels, describing rigorous annotation protocols on legal-domain text and lexicon-supported methods [17]. Cross-lingual resources spanning the region have also emerged: for example, a professionally curated Uzbek–Kazakh parallel corpus assembled from multi-genre sources (literature, news, UDHR) provides a foundation for MT and cross-lingual transfer among Turkic languages [12]. Together, these resources address structural challenges of agglutinative morphology and limited corpora, and they enable benchmarking of wider model families on Central Asian languages.

Transformer methods for text classification in low-resource settings have been propelled by multilingual pretraining. XLM-R, trained on CC-100 across 100 languages, shows strong cross-lingual transfer on low-resource benchmarks, illustrating the impact of scale and data quality despite the curse of multilinguality [1]. Industrial evaluations show that multilingual encoders can outperform monolingual models in certain classification settings while domain-adaptive pretraining and simple augmentation yield gains without additional labeled data [2]. In low-resource settings, zero-shot transfer from high-resource languages has proven effective, highlighting the value of cross-lingual transfer when labeled target data are scarce [3]. These findings align with Uzbek-specific model development, where the availability of monolingual resources and benchmarks has enabled stronger language models and downstream evaluation [10], [12], [17].

MTL has emerged as a principled mechanism to share supervision across related tasks, often improving generalization and sample efficiency. Prior surveys review a wide range of multitask learning architectures and training strategies, distinguishing auxiliary-task and joint MTL formulations and highlighting optimization techniques crucial for stable training with pretrained language models [4], [6]. Empirical studies confirm the value of shared encoders and task-related sampling: Rotman and Reichart introduce multi-task active learning for pre-trained Transformers and demonstrate that carefully coordinated annotation across related tasks (e.g., NER with dependency parsing or slot-filling) reduces labeling cost and improves target performance [9]. Such MTL patterns—shared encoders with lightweight task heads and auxiliary linguistic supervision—are particularly relevant for Central Asian languages, where labeled data are scarce and shared signals benefit multiple NLP tasks. [2], [10], [17].

While most prior work differs in task formulation or data scale, several studies have explored multitask learning for document-level news classification. A closely related work is the Czech News Classification (CZE-NEC) dataset by Kydlíček and Straka (2023), which studies multitask learning for document-level news classification. [18]. Other researches have examined individual components of our problem setting. Multi-task text classification has been studied for efficient topic prediction using shared encoders [5]. Temporal attributes

TABLE I: Label distribution in the Kun.uz dataset.

Attribute	Category	Count	%
Gender	Male	14,798	91.60
	Female	1,356	8.40
Year	2021	2,972	18.40
	2022	3,441	21.30
	2023	3,460	21.42
	2024	3,189	19.74
	2025	3,092	19.14
Topic	Uzbekistan	7,176	44.45
	International News	5,029	31.14
	Society	1,767	10.94
	Sport	1,083	6.71
	Economics	854	5.29
	Technology	138	0.86
	Other	107	0.66

such as publication time have been studied through document-level temporality modeling [14]. Demographic inference from blog posts (gender, age, industry), has been explored using multitask learning [15]. An initial benchmark for Uzbek text classification has been proposed, addressing topic classification and offering preliminary analysis in a low-resource setting [7]. While these studies address topical, temporal, and demographic prediction separately, they are typically investigated in isolation and outside a unified news classification framework. In contrast, our work brings these complementary tasks together in a single multitask formulation for document-level Uzbek news classification.

III. DATASET

A. Corpus Description

We introduce a new Uzbek news dataset collected from Kun.uz, one of the most widely read online news platforms in Uzbekistan. The dataset includes article texts along with associated metadata such as author names, publication dates, topic labels, extracted from the official website and Telegram channel. In total, the dataset comprises 16,154 news articles containing approximately 8 million words, written by 63 authors, with an average of 256 articles and around 125,000 words per author. Topic tags and links are also extracted and preserved for potential use in future research. The dataset is constructed to support NLP research on Uzbek news text. We plan to release a cleaned and extended version of the dataset in future work.

B. Annotation and Variables

The dataset is relatively clean, as it is collected from an official news source. We use three labels in this study: topic, publication year, and author gender (Table I). Author gender is inferred from common Uzbek surname endings (e.g., *-v* for male and *-va* for female) which is highly predictive in Uzbek and Russian naming systems [11]. Ambiguous cases are manually corrected, while the majority of names follow the expected morphological patterns. Topic labels are taken

directly from the news categories, and publication year was derived from the article date.

C. Preprocessing Pipeline

We applied minimal preprocessing pipeline focused on data quality. We deliberately avoid artificial class balancing, as it would distort the natural distribution of Uzbek news topics and reduce ecological validity. Models are therefore trained and evaluated on data that reflect realistic deployment conditions. During data extraction, a small number of empty or noisy records are identified and removed. Records with empty article text are removed, and entries with missing author information are filtered out. We derive structured labels from metadata (topic from category, year from publication date, and gender inferred from surname endings). The original 17 topic categories are merged into 7 by grouping low-frequency categories into an *Other* class. Apart from basic text normalization, we retain the original news text without aggressive preprocessing.

IV. METHODOLOGY

A. Single task models

For classical baselines, namely Support Vector Machines (SVM) and Logistic Regression (LR), we represent texts using a TF-IDF vectorizer with unigrams and bigrams, excluding tokens appearing in fewer than two documents or in more than 90% of documents. These single-task baselines are trained independently for each task using cross-entropy loss. We include a bilingual LSTM recurrent network in which input texts are padded to 512 tokens and represented using 330-dimensional trainable embeddings. The model consists of a bidirectional LSTM with 128 hidden units, followed by a fully connected layer and a task-specific output layer with softmax (multi-class) or sigmoid (binary) activation. Optimization is performed using AdamW with a weight decay of 0.01 and a linear learning-rate schedule with 10% warmup, using a learning rate of $2e-5$.

B. MTL models

We introduce multi-task learning variants to evaluate the effectiveness of shared representations for gender, publication year, and topic prediction. The variants are designed to systematically compare input processing strategies within an otherwise identical model architecture. We employ BERT_{bek}-News-Big (hereafter BERT_{bek}), a pretrained transformer model for Uzbek news text, as the backbone of our experiments. The news articles in our dataset are relatively long, with a mean length of 488 words. Given the constraints of transformer models, inputs are tokenized and truncated to a maximum of 512 subword tokens, corresponding to the model’s maximum supported input length, allowing us to retain most of the article content. We also evaluate reduced multitask configurations (e.g., topic+year) to analyze how including gender prediction affects shared representations and task interactions. The multi-task learning (MTL) model is trained using the same optimization settings as the LSTM. Our standard multitask model consists of a single shared

encoder followed by task-specific classification heads, trained jointly across all tasks. We evaluate an enhanced multitask variant incorporating cross-task attention mechanisms, along with adaptive loss weighting through dynamic task weighting and class-weighted cross-entropy, to better balance task contributions during training. We also experiment with a layer-wise multitask variant that aggregates representations from the final four encoder layers before task-specific prediction, motivated by findings in BERT that the top four hidden layers of the pre-trained Transformer capture the most task-relevant semantic information [13]. We optimize MTL models using task-specific prediction losses under a unified training objective. For the standard multitask model, the total loss is defined as the sum of the three task-specific prediction losses:

$$\mathcal{L} = \mathcal{L}_g + \mathcal{L}_a + \mathcal{L}_i$$

In the layer-wise multitask variant, representations from the final four encoder layers are combined using learned weights and the same joint loss as the standard multitask model is used. For the cross-attention multitask variant, task losses are combined using learned uncertainty parameters to balance task contributions during training. [8]:

$$\mathcal{L} = \sum_{k \in \{g, y, t\}} \left(\frac{1}{2\sigma_k^2} \mathcal{L}_k + \log \sigma_k \right) \quad (1)$$

All models are trained using the same setup and input processing to ensure fair comparison across variants. We use different batch sizes across models depending on their computational cost. The dataset is split into 85% training and 15% test sets, stratified by gender to preserve distributions. Model performance is evaluated using Accuracy and Macro-averaged F1 (Macro-F1) across all tasks. Macro-F1 is reported to account for class imbalance and to ensure equal weighting of all classes. Training is performed using PyTorch on cloud based GPUs provided by Kaggle and Google Colab.

V. RESULTS OF EXPERIMENTS

Table II and Fig. 1 report accuracy and macro-averaged F1 (F1) for the three prediction tasks: author gender (G), publication year (Y), and topic (T).

Overall, transformer-based models consistently outperform classical and recurrent baselines, with BERT_{bek} variants achieving the strongest performance across tasks. For gender classification, all top-performing systems reach high accuracy (0.93–0.94) and macro-F1 (0.77), indicating that gender-related cues are encoded in the text and can be effectively captured by both TF-IDF-based linear models and BERT_{bek} fine-tuning. In contrast, publication year prediction is substantially more challenging: classical baselines achieve only moderate performance (e.g., SVM with 0.64 accuracy and 0.64 macro-F1), while the LSTM baseline performs poorly (0.21/0.18), suggesting that long-range contextual information and strong pretrained representations are crucial for temporal inference. BERT_{bek} Year-Topic MTL achieves the best year prediction performance (0.66–0.67), indicating that excluding

TABLE II: Performance of Single-Task and Multi-Task Models on the News Dataset

Model	G Acc	G F1	Y Acc	Y F1	T Acc	T F1
LR	0.86	0.58	0.47	0.46	0.73	0.54
SVM	0.94	0.77	0.64	0.64	0.83	0.67
LSTM	0.90	0.49	0.21	0.18	0.25	0.20
BERTbek (Standard)	0.94	0.77	0.65	0.66	0.83	0.69
BERTbek (CA-MTL)	0.93	0.77	0.65	0.66	0.83	0.70
BERTbek (Layer Wise)	0.94	0.77	0.64	0.65	0.82	0.68
BERTbek (Year-Topic)			0.66	0.67	0.84	0.69

gender leads to more effective year prediction than including it. For topic classification, BERTbek Cross-Attention MTL performs best, reaching up to 0.70 macro-F1, while the BERTbek Year+Topic configuration achieves the highest topic accuracy, suggesting that inclusion of gender prediction may introduce task interference for this task. The layer-wise aggregation MTL variant does not provide consistent gains, matching the strongest baselines on gender (0.94/0.77) but slightly underperforming on year and topic.

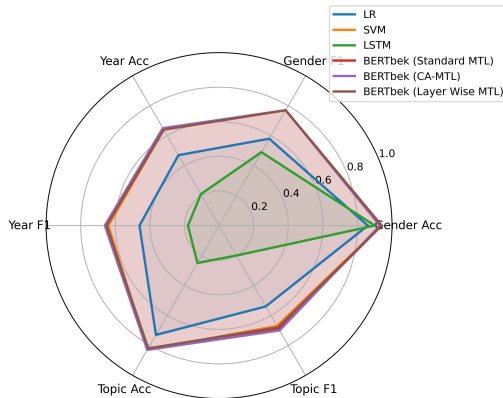


Fig. 1: Multitask Classification Results on the News Dataset.

VI. DISCUSSION

VII. ANALYSIS

A. Task Interactions and Transfer Effects

Task interactions across gender, publication year, and topic classification are generally limited. Modest positive transfer is observed between topic and publication year, likely due to

shared event-related lexical patterns, while gender classification remains largely independent and shows minimal sensitivity to joint training. Additional experiments assessing the contribution of gender to the multitask learning setup indicate that including gender slightly degrades overall performance, likely due to its high class imbalance. This indicates that including weakly related and highly imbalanced tasks in a multitask setting may be detrimental, and that careful task selection is crucial for effective multitask learning.

B. Error Analysis

Figure 2 provides a detailed view of prediction errors through confusion matrices for the three tasks—gender, publication year, and topic—using the best-performing BERTbek Cross Attention MTL model. Confusion matrix analysis shows distinct error patterns across tasks. Although gender classification achieves high overall accuracy, the corresponding macro-F1 score is lower due to class imbalance. Recall for the female class is reduced, reflecting the dominance of male-authored articles in the dataset.

Publication year prediction performs reasonably well, revealing a strong temporal proximity bias: 77.3% of misclassifications occur within ± 1 year of the true label. Specifically, adjacent-year errors (2021-2022, 2022-2023, etc.) account for the vast majority of mistakes, with errors decreasing exponentially as temporal distance increases (± 2 years: 18%, ± 3 years: 4%, ± 4 years: 1%). This pattern reflects gradual temporal drift in language use and indicates that the model captures a continuous temporal signal despite being trained with a categorical objective.

For industry classification, the result reflects pronounced class imbalance: majority categories (e.g., *Sport* and *Uzbekistan*) show strong diagonal dominance, while minority classes tend to have lower recall and confusion with related topics, showing the gap between accuracy and macro-F1.

C. Ablation Studies

We conduct an ablation study comparing the standard BERTbek multi-task model with variants incorporating layer-wise aggregation and cross-attention. Across all tasks, performance differences between variants are modest, with no component providing consistent improvements. Layer-wise aggregation can potentially capture complementary features across transformer layers, but in our experiments it does not improve performance. This is likely because the gender task generates a consistently strong signal in the shared encoder, which dominates the learned representations and limits the benefit of combining information from multiple layers for subtler semantic tasks such as publication year and topic prediction. Compared to standard multitask learning, the cross-attention variant provides marginal gains, particularly for topic classification, suggesting that cross-attention mechanisms combined with dynamic task weighting are beneficial in our multitask setting. We also observe that the limited gains from the Cross-Attention and Layer-wise variants are likely due to the relatively small dataset size and the simplicity of news language.

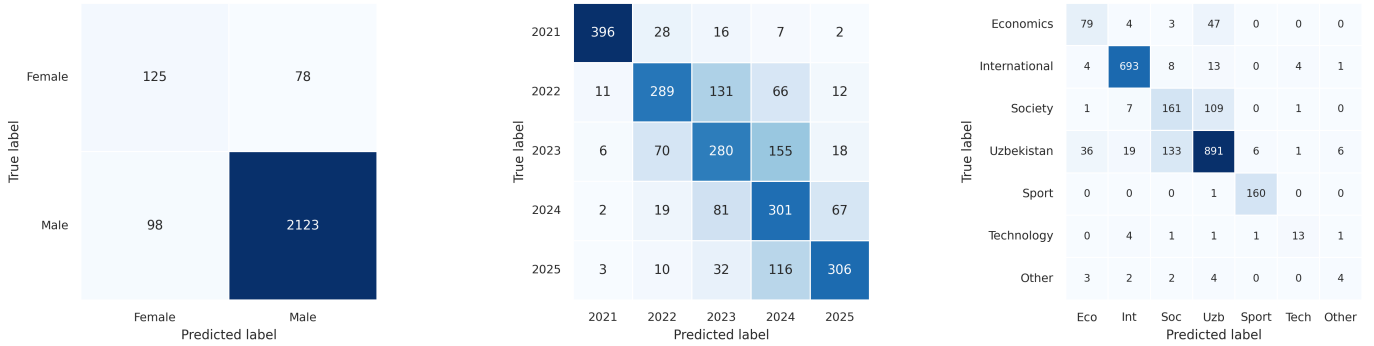


Fig. 2: Confusion matrices

Given the characteristics of our dataset, the additional model complexity introduced by the Cross-Attention and Layer-wise architectures may not provide meaningful benefits

D. Relation to Representation Structure

To enable direct comparison across tasks, UMAP is computed once on the shared sentence representations, and the same projection is visualized with different label colorings, as shown in Fig. 3. The structure is shared across all visualizations, as it is derived from a single multi-task encoder and mainly driven by semantic content. Gender labels demonstrate clear separation, consistent with near-saturated performance. Although topic labels are noisy and overlap, they partially align with the dominant semantic organization of the representation, whereas publication year shows weak alignment, leading to diffuse and overlapping patterns.

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion of key findings

This study demonstrates that a shared multitask encoder can learn representations that are informative for publication year and topic classification, and moderately useful for gender prediction in Uzbek news, capturing both task-specific and shared patterns while supporting joint training. Gender and topic benefit from stable stylistic and semantic cues, while publication year remains more challenging due to temporal variation in language. In our setting, task performance is mostly determined by how well task labels align with the learned shared representations, rather than by increased architectural complexity or additional modeling components. Ablation and training dynamics further indicate diminishing returns from additional components, suggesting a ceiling effect driven by dataset characteristics.

B. Limitations

We acknowledge several limitations that should be considered when interpreting our findings. The findings are based on a single Uzbek news dataset and a fixed experimental setup. Therefore, conclusions regarding task behavior, representation structure, and performance trends may not generalize to other languages or datasets. Model performance saturates early,

indicating a ceiling effect that is likely driven by limited and diffuse task-specific signals in the data. Topic labels shows class imbalance and semantic overlap, which affects minority-class recall and constrains macro-F1 despite strong overall accuracy. We use a maximum sequence length of 512 tokens, while the mean document length in our dataset is 488 tokens. As a result, longer documents may be truncated, potentially limiting the model’s ability to fully capture their representations and slightly affecting performance. We view these limitations as opportunities for future research beyond the scope of this study.

C. Contributions to the field

This work presents a systematic multi-task learning study on Uzbek news classification, jointly predicting author gender, publication year, and topic using BERTbek-based models. Beyond reporting competitive performance, we provide in-depth analysis with confusion matrices and shared UMAP projections, revealing how task labels align with the learned semantic representation. Gender and topic prediction rely on relatively stable stylistic and semantic cues, whereas publication year prediction is constrained by weaker temporal signals, leading to saturated performance across model variants. These findings provide empirical insights into representation structure, task difficulty, and performance ceilings in multitask learning for low-resource languages, while also showing that pretrained representations support robust performance across multiple tasks.

D. Future Work

This study suggests several directions for future work. Future work can explore finer-grained temporal modeling approaches, such as ordinal or continuous formulations, to better capture gradual linguistic change over time. Expanding and rebalancing the dataset may help mitigate label noise and class imbalance, particularly for gender and topic classification. Evaluating the approach on other news corpora and languages would help clarify the generalizability of our findings. Given the length of news articles and the use of the maximum 512-token input window in this study, future work may investigate multilingual long-context transformers that can more fully exploit document-level information. These

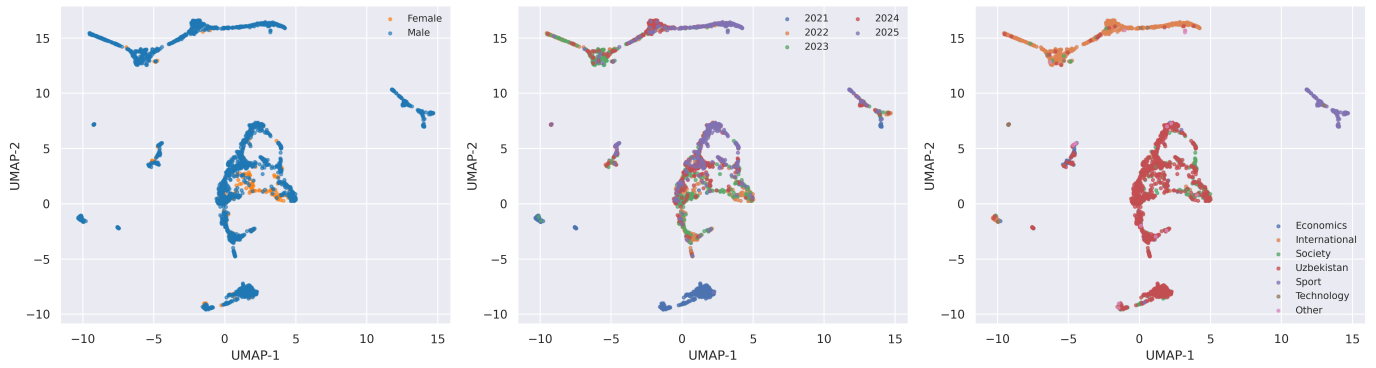


Fig. 3: UMAP 2D projections

directions provide opportunities to extend the current findings and improve generalization.

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